

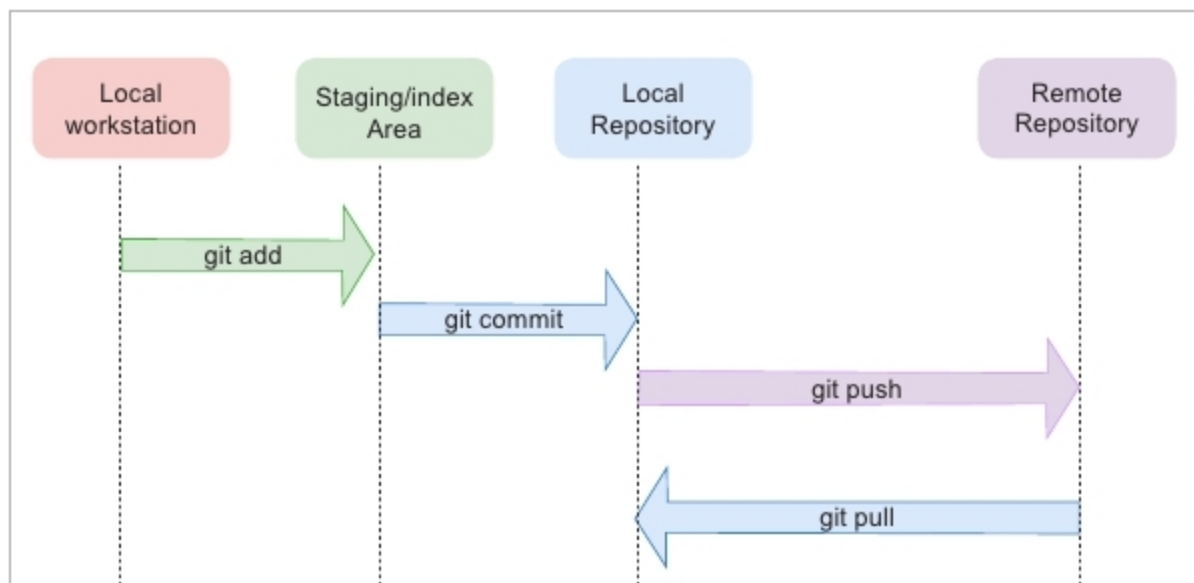


Figure 6: The Git workflow

Git has two repositories — local and remote. The local repository sits on the developer's personal system and has the complete codebase. Developers can make and commit any changes they want in their local codebase.

Files can be either tracked or untracked in the working directory. A newly created file is initially always in the untracked state, which means the changes made in that file are still not tracked by Git. Tracked files are the ones present in the previous commit. They can be in the following three states.

- *Modified* means the file has been changed but hasn't been committed yet.
- *Staged* means a modified file is marked in its current version to go in the next commit snapshot. The file is moved to the staging area.
- *Committed* means that the data is safely stored in the database.

To reflect these changes in the global codebase, developers need to push them to a remote repository, which is usually hosted on Web based file hosting services like GitHub, GitLab, etc. A remote repository is where developers can show their commits and collaborate with the whole team. Also, if they want the changes made by other developers to reflect on their local repository, they can pull the code from a remote repository. The complete workflow is shown in Figure 6.

Installation

Check if your machine already has Git by typing the following command at the terminal:

```
$git --version
```

If the above command does not return the Git version, then take the following steps to install Git.

1. Linux

For Ubuntu/Debian, type:

```
$sudo apt-get update
$sudo apt-get install git
```

For Fedora, give the following command:

```
$sudo dnf install git
```

2. Windows

- Download the latest installer from <https://git-scm.com/download/win>.
- Go through with the installation wizard. It's a good idea to keep the default options.

3. MacOS

Download the installer from <https://git-scm.com/download/mac> and go through the prompt. Or, install via *homebrew*, using the following command:

```
$brew install git
```

Configuration

The first most important step after

installation is to configure the system for things like setting up identity, as it will be embedded in every Git commit done. Once set, these *configs* stick around almost forever.

```
$git config --global user.name "<your name>"
$git config --global user.email "<your email ID>"
```

Setting up a Git repository

There are two ways to set up a Git repository.

1. Clone a Git repository from a source on the Internet. In other words, it creates a local copy of a remote repository in the machine, which is shown in Figure 7.
2. Take a local directory (empty or non-empty) and initialise it to become a Git repository, as follows:

```
$mkdir git-demo; cd git-demo
$git init
```

Recording changes

Let's create a new file called *file_1.txt* in the previously initialised *git-demo* directory and put some content into it to see how changes are tracked and committed in the Git repository. *file_1.txt* is in the working directory but not yet tracked by Git. In order for Git to

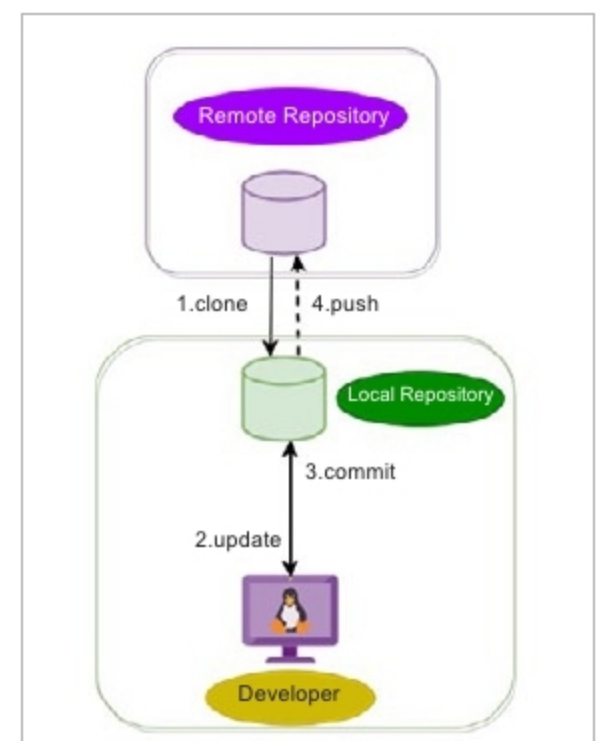


Figure 7: Flow diagram of a Git clone